Re: Some questions on your paper regarding the academic performance of East Asian schoolchildren (1967-2020)

1 message

Aryan T <athanki5670@gmail.com> 6 May 2024 at 01:28
To: tompiccolo50@gmail.com

Sent from my iPhone

On 21 Oct 2023, at 09:13, Björn Boman <bjorn.boman@edu.su.se> wrote:

Hello Aryan

Likewise. Thanks for the questions.

I have some interesting key points to deliver. First of all, I think we should set Singapore aside for a moment. Only 75% of Singaporeans are Chinese and the country is characterized by selective migration, much like Beijing and Shanghai.

It is also better to set aside PISA scores (whether the samples are representative) for a moment and look at IQ scores, which are largely genetic within populations. If we compare WISC-III scores, we notice that South Koreans do not have higher full-scale IQ scores than Swedes or Germans (compare for example Kwak, and Ramund and Sonnander in Culture and Children's Intelligence. Cross-Cultural Analysis of the WISC-III).

We could also use Central Asian populations as a line of indirect evidence against the genetic hypothesis. Kazahks and Uzbeks are predominantly Mongolian populations, just like the Japaense and Koreans. Yet, Kazakhs have relatively low IQ scores and PISA scores (about 0.8-1.2 of an international standard deviation, depending on which countries that are being compared). In the same country context, the Russians outperform both of these genetically East Asian populations. That in turn indicates that there are economic (income inequality, GDP per capita), cultural (diligence - Kazakhs are mostly Muslims, not Confucians, Christians or Westerns regarding their cultural proclivities) and educational factors that explain these differences in cognitive skills (IQ, PISA).

My articles of relevance:

Boman, B. (2020). What makes Estonia and Singapore so good? Globalisation, Societies, and Education. https://www.tandfonline.com/doi/abs/10.1080/14767724.2019.1701420

Boman, B. (2023). Is the SES and academic achievement relationship mediated by cognitive ability? Evidence from PISA 2018 using data from 77 countries. Frontiers in Psychology, 14.

https://www.frontiersin.org/articles/10.3389/fpsyg.2023.1045568/full

Boman, B. (2023). The influence of SES, cognitive, and non-cognitive abilities on grades: cross-sectional and longitudinal evidence from two Swedish cohorts, European Journal of Psychology of Education, 38.

https://link.springer.com/article/10.1007/s10212-022-00626-9

Boman, B. Mosesson, M. (2023). Converging cultures? A comparative analysis of South Korea and Sweden's national curricula, 1980–2018. Discover Education.

https://link.springer.com/article/10.1007/s10212-022-00626-9

Other references of importance:

Grigoriev, A. Lynn, R. (2014). A study of the intelligence of Kazakhs, Russians and Uzbeks in Kazakhstan. Intelligence, 46(September–October), 40-46

Kwak, K. (2003). South Korea, in Georgas (2003). in Culture and Children's Intelligence. Cross-Cultural Analysis of the WISC-III. (eds.) J. Weiss, L. Vijervan de, and D.F. Saklofske (Amsterdam, Netherlands: Elsevier)

Lynn, R. H-Y. Chen, Y-H. Chen. (2011). Intelligence in Taiwan: Progressive matrices means and sex differences in means and variances for 6–17 year olds. Journal of Biosocial Science, 35(4), 469-474.

Best wishes,
Björn

Från: Aryan T <athanki5670@gmail.com>
Skickat: den 19 oktober 2023 01:56
Till: Björn Boman
Ämne: Re: Some questions on your paper regarding the academic performance of East Asian schoolchildren (1967-2020)

Hello

I hope you are well.

I have listed some observations and questions, I have numbered these and am hopeful you could answer them or comment on them in turn.

1) You mentioned that all countries could improve, but also implied that the extent of improvement would differ from country to country even if all had, for example, the best education, the best diligence, etc. Why? (You said that by how much scores would improve is "another question")

2) Could all countries be on par with Shanghai or Singapore? You said that group differences (I.e between countries) are more environmental. Does that mean that the reason that in Shanghai, the students are top performers (Level 5-6) in both Maths and Science is mostly due to nurture? This is not the case in any other PISA region examined.

3) Canada has been found in research to submit unrepresentative samples. The same has been found for Switzerland, which has a high student exclusion rate. Switzerland's score for cycles where it submitted a representative sample was markedly lower than say, its performance in PISA 2012. I also understand there have been issues with other countries such as Finland submitting unrepresentative samples.

4) South Korea also from 2012 to 2015 increased its participation rate (to 92%- a level not seen before. Its prior participation rates were all lower than this) and its score dropped to 524.
A score close to 524 is the same score that England would achieve if everyone studied intellectually demanding material at GCSE (had a much better curriculum) AND had the best education, the best resources, the best or highest levels of diligence. This is according to research that says England could be close to South Korea's performance.

5) This, interestingly does not correspond with over 30% of England's students at levels 5-6 in Maths, but maybe 17%. This 30% figure was the result for South Korea in PISA 2012, which is clearly unrepresentative. So clearly there are underlying issues with potential bias or misrepresentation.

6) If South Korea was to submit a representative sample, what would its scores for Maths and Science look like?

7) Also, could you name the specific regions to which intelligence (rather than nurture) would explain group differences- why these places do better.

8) Taiwan's scores for Maths are higher than the samples for Canada and Estonia (of which Canada is an unrepresentative sample as found by research)

9) What did you mean by:

"But, overall, yes it is the case"- last paragraph. "Yes, it is the case" to what? I didn't quite understand.

"It seems that some East Asian countries are not fully maximising their potential"- for this statement, the performance of countries like the UK would only just match countries like South Korea if the full potential of all teachers and hence students was expressed. The UK or England could be 5th in Maths (according to PISA 2009) if the full potential of all students was to be exploited, according to other research.

10) The UK being 5th in Maths would still not constitute academic excellence (e.g all A*s) for all. Is this evidence intelligence/innate ability/genetic factors has a significant bearing on academic performance?

11) Clearly, there are issues with participation rates for several countries. What would the scores look like if fully representative samples were to be chosen?

12) Are there any issues with participation rates or perhaps potential bias for any countries in TIMSS?

I would also appreciate it if you could link the papers you mentioned in your reply.

Many thanks

Aryan Thanki

On Thu, 24 Aug 2023 at 16:48, Björn Boman <bjorn.boman@edu.su.se> wrote:

> Dear Aryan,
>
> Thank you for this e-mail.
>
> Some of these questions are answered within the review article and some are beyond the scope of the article. I summarize a few key points.
>
> 1. Regarding China, B-J-S-Z, this sample is quite unrepresentative. Other countries in East Asia are much less so.
>
> 2. I am not saying that nurture is the reason why individuals are outperforming others, but groups on the national level. Those are two different things. It is possible that individual genetics explain 65% of the variance in academic achievement as IQ is largely genetic, while group differences are perhaps 0-10% genetic. However, this topic does partly transcend the scope of my paper.
>
> 3. Selective migration is mentioned in the discussion section of the paper. It applies to for example Singapore and Shanghai.
>
> 4. I agree that average and low performers within e.g. PISA and TIMSS could improve their scores substantially if they had better teachers, curricula, classroom climates, resources, higher acerage SES, more diligence etc. How much is another question. But please note that Estonia and Canada are on par with South Korea and Taiwan. Not every East Asian country maximizes its potential, it seems. But overall, yes, it is the case.
>
> Ps. If you are interested, I could send you more links to my papers on Singapore/Estonia, IQ/PISA etc. Or you can just Google.
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> Best wishes,
> Björn
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> Från: Aryan T <athanki5670@gmail.com>
> Skickat: den 24 augusti 2023 00:35:28
> Till: Björn Boman
> Ämne: Some questions on your paper regarding the academic performance of East Asian schoolchildren (1967-2020)
>
> Hello
>
> I hope you are well.
>
> I had some questions regarding your paper, the 1967-2020 analysis of all literature concerning the academic success of East Asian schoolchildren.
>
> Firstly, is it true that for the top performers in countries such as South Korea and Japan, those on levels 5-6 in Maths and/or Science in PISA- nurture is the single explanatory factor? Do they do well simply because they are pushed and work hard? Does this imply that the majority of students can reach these high levels of attainment- as anyone can be pushed and work hard?
>
> Does this not contradict established research suggesting nature is more important than nurture for academic success? For GCSE examination results in Maths and Science in particular, studies have found that nature matters more than nurture regarding academic excellence on an individual level.
>
> Or, rather than contradicting the research, does it suggest that countries in the West such as the United Kingdom are not currently fully exploiting the potential of their students- and that more students (but not all, or even the majority) are capable of reaching standards of academic excellence in their GCSE examinations?
>
> I ask this as research has been carried out to show what would happen if all students in the UK, as of PISA 2015, adopted the mindset and principles of East Asian countries as well as studying more intellectually demanding material. The UK's PISA Maths score would rise to 524- between a grade 5 (C) and 6 (B) using the 9-1 GCSE system- a high grade 5 (C).
>
> Other research shows that if the full potential of all teachers and therefore students was expressed, the UK would rise to the top 5 in PISA Maths, so the students would be averaging slightly higher GCSE grades on average.
>
> These scenarios are obviously not reflective of an average academically excellent standard or norm, but shows more people are capable of reaching high standards of performance in education.
>
> Rather than contradicting established research on how academic achievement relies on joint effects of nature and nurture, does the high academic achievement of East Asian students simply mirror that these countries are to a greater degree than the West, exploiting the full potential of their students? That more students are capable of performing at high levels than currently being observed.
>
> Therefore- does it still mean that the achievements of those at the top is still heavily dependent on both intelligence/academic ability and excellent nurture? Given that increasing equity to education would increase heritability and the effect of genetic differences.
>
> Also, would this grade increase affect all groups equally? What about those ethnic groups in the UK- Indian and Chinese- who already have this higher level of achievement? For these groups is it likely that their full potential as a group has been reached, due to the extra focused hard work and dedication they put into their education?
>
> I believe that to say the achievement of students at the top entirely being attributed to nurture would be pseudoscientific in nature.
>
> There is one non selective school in the UK known as Michaela Community School (www.michaela.education) which obtains grades 6 and 7's for its students at GCSE, on average.
>
> This is clearly higher than estimates for the average grade increase if all students' potential was exploited and educational equity to be achieved nationally (as mentioned earlier)
>
> I must stress that the disciplinary climate and standards of teaching instruction in this school seems to be similar to the standards in East Asian countries such as South Korea.
>
> Despite this, average students in South Korea or other East Asian countries are not performing at the same level of these students- note that that the disciplinary climate and standards of teaching instruction in this school seems to be similar to the standards in these East Asian countries. Why is this?
>
> It must be noted that the school in question does require you to actively apply, though.
>
> My question is, that if there are predictions or estimates on what would happen to a country's average PISA score if the full potential of all the teachers (or students) was exploited, leading to a given increase in average PISA scores- but there are examples of non selective schools in that country that have achievements higher than the estimations (these estimates are based on very large population sizes, thousands or tens of thousands)- what does this mean? Could it be assumed those schools (which have a small sample size relative to the estimation sample size) are not representative of the entire population in that country?
>
> Furthermore, regarding the academic excellence seen in places such as Singapore, Taiwan and Shanghai, what factors cause this? Is selection and selective migration a factor?
>
> For Asian Americans, they are selected for- does this have an impact on their academic success? Would it be reasonable to say intelligence or differences in intelligence within the group matter? Intelligence was found to explain 58% of the variance within Asian Americans (at a maximum) of individual differences up to grade 10 (GCSE-AS level for the UK), with this dropping to 25% after grade 10. Why is this?
>
> Also, regarding the increase in PISA scores from BSJG (2015) to BSJZ in 2018's PISA- Shanghai took part in PISA 2012 and scored 612, with 55% at level 5 or 6 (30% Level 6) in Mathematics. Is it possible that Jiangsu and/or Guangdong were over represented in PISA 2015 leading to lower observed scores? Is it possible that the "cultural and social" variables you mentioned which lead to the massive increases in PISA scores was simply a result of the full potential of students in regions such as Jiangsu and Beijing (which experience similar selective migration to Shanghai) being exploited?
>
> Therefore- their high attainment not representative of what other countries could attain?
>
> Is the theoretical maximum PISA score for all OECD countries the same? Does each country possess the same level of intellectual potential, with current differences being entirely environmental? I ask as a paper exists in the reference list to the 1967-2020 analysis on an "expert opinion on international differences in cognitive ability (PISA) test results"- differences in genetics were given as the second most important factor, second to environmental factors being the most important factor (>53% variance explained)
>
> Does this mean that while each country can definitely improve its performance in PISA significantly, the extent of improvement will vary due to genetic factors?
>
> Many thanks and I hope to receive a response in due course-
>
> Aryan Thanki